Beyond Monolingual Boundaries: Sarcasm Detection on Code-Switched Hinglish Social Media Text

Jaideep Manupati
jaideepm21@vt.edu
Virginia Tech
Alexandria, Virginia, USA

Akshada Rane
akshada@vt.edu
Virginia Tech
Alexandria, Virginia, USA

Gokulakrishnan Ravi
gravi@vt.edu
Virginia Tech
Alexandria, Virginia, USA

Abstract

Code-switching – alternating between two or more languages within a single utterance is pervasive among Hindi-English (Hinglish) speakers on platforms such as Twitter and Reddit. Sarcasm detection in this setting compounds the challenge: it requires understanding figurative language, cultural context, and pragmatic intent, even as standard NLP tokenizers and monolingual training pipelines break down on code-switched input. This paper systematically compares three paradigms for binary sarcasm detection on Hinglish code-switched text: (1) supervised fine-tuning of **MuRIL** [4], a multilingual encoder pre-trained on Indian languages; (2) zero-shot prompting of **LLaMA-4 Scout 17B** [5] via the Groq API; and (3) cross-lingual transfer from English to Hinglish using **XLM-RoBERTa-large** [2] trained on iSarcasmEval [3]. Evaluated on a held-out 10% stratified test split of the Swami et al. [6] Hinglish Sarcasm Dataset (525 tweets), MuRIL achieves the best Macro F1 of **0.8683**, followed by LLaMA-4 at **0.7660**. XLM-RoBERTa attains high accuracy (0.9048) but a Macro F1 of only **0.4750**, indicating near-total failure to transfer sarcasm detection knowledge across language pairs. Prompt calibration accounts for ≈40 F1 points in zero-shot LLM evaluation, with a single base-rate-naive prompt collapsing performance below the majority-class baseline.

Keywords

sarcasm detection, Hinglish, code-switching, MuRIL, LLaMA-4, XLM-RoBERTa, cross-lingual transfer, zero-shot prompting, script-level analysis

1 Introduction

Code-switching — the practice of alternating between two or more languages within the same conversation, sentence, or phrase — is one of the most natural communication phenomena among bilingual speakers, yet one of the most challenging for NLP systems. In India, where Hindi and English are both widely spoken, "Hinglish" has become the dominant register of online communication on platforms such as Twitter and Reddit. A tweet such as *"Waah, kya service hai, bahut impressed!"* is immediately recognizable to any bilingual reader as sarcastic mockery of poor service — yet standard NLP systems routinely classify it as positive sentiment. The difficulty is threefold. First, code-switching breaks the assumptions of standard tokenizers and vocabulary sets. Second, sarcasm requires inference about the speaker's pragmatic intent — what they *mean* versus what they *say*. Third, Hinglish is primarily written in Roman script in our dataset. Romanized Hindi and English often appear within the same post, creating orthographic variation that challenges models trained on clean monolingual text. This project investigates three paradigms for Hinglish sarcasm detection:

- **Supervised fine-tuning (Approach 1):** Fine-tuning MuRIL [4], a BERT-based encoder pre-trained on 17 Indian languages and their transliterations, with class-weighted cross-entropy to address the 9:1 label imbalance.
- **Zero-shot prompting (Approach 2):** Querying LLaMA-4 Scout 17B [5] through the Groq API with a carefully calibrated prompt, without any task-specific training data.
- **Cross-lingual transfer (Approach 3):** Training XLM-RoBERTa-large [2] on English sarcasm data from iSarcasmEval [3] and evaluating zero-shot on Hinglish.

Our results show that supervised fine-tuning on in-domain data remains the strongest approach (Macro F1 = 0.8683), but a well-calibrated LLM achieves competitive zero-shot performance (Macro F1 = 0.7660). Cross-lingual transfer fails near-completely (Macro F1 = 0.4750) — a result that is itself informative about the limits of figurative language transfer on code-switched text.

2 Related Work

Code-switching benchmarks. Aguilar et al. [1] introduced LinCE, a centralized benchmark for linguistic code-switching evaluation, highlighting that models trained on monolingual corpora consistently underperform due to intra-sentential mixing, script variation, and informal orthography.

Hinglish sarcasm. Swami et al. [6] introduced the Hinglish Sarcasm Dataset, a corpus of 5,250 English-Hindi code-mixed tweets with token-level language identification annotations covering Roman-script and transliterated Hindi.

Multilingual pre-training for Indian languages. Khanuja et al. [4] introduced MuRIL, a BERT-based model pre-trained on 17 Indian languages and their transliterations, designed to handle the romanization patterns common in Hinglish text.

Cross-lingual transfer. Conneau et al. [2] demonstrated that XLM-RoBERTa achieves strong zero-shot cross-lingual transfer on syntactic and semantic classification tasks, but their evaluation leaves open the question of figurative language transfer in code-switched settings — the gap our Approach 3 directly probes.

LLMs for multilingual tasks. Lavine et al. [**?**] extended instruction-following to 100 languages with xLLMs-100. Our zero-shot LLaMA-4 experiment [5] targets the gap between multilingual LLM capability and naturally mixed-language pragmatic inference.

Table 1: Dataset split statistics (stratified, `random_state=42`). Identical split across all three approaches.

Split	Size	Sarc.	Non-S.	Rate
Train (80%)	4,200	403	3,797	9.6%
Val. (10%)	525	51	474	9.7%
Test (10%)	525	52	473	9.9%

Table 2: Approach-specific preprocessing decisions.

App.	Model	Preprocessing
1	MuRIL	Remove noise/URLs. **Standardize Romanized Hinglish**, Hinglish casing, punctuation.
2	LLaMA-4	Remove URLs. Preserve Roman Hinglish, emoji, punctuation.
3	XLM-R	Strict symmetry with iSarcasmEval. Strip leakage tags (#sarcasm, #irony).

3 Dataset

3.1 Hinglish Sarcasm Dataset

Our primary dataset is the Hinglish Sarcasm Detection Dataset [6], containing 5,250 English-Hindi code-switched tweets annotated with binary Sarcastic/Non-Sarcastic labels. The dataset includes token-level language identification annotations tagging each word as English (`en`), Hindi (`hi`), or Other (`rest`, covering hashtags, mentions, and punctuation). The raw dataset is heavily imbalanced: 504 sarcastic (9.6%) versus 4,746 non-sarcastic (90.4%). We apply a stratified 80/10/10 train/validation/test split with `random_state=42` uniformly across all three approaches, ensuring all models are evaluated on the exact same held-out test set for direct comparison. Table 1 summarizes the split statistics.

3.2 iSarcasmEval

For Approach 3, we additionally use iSarcasmEval [3], an English sarcasm benchmark with author-annotated labels. We use Task A (binary sarcasm classification) for primary training and Task C (sarcastic/literal rephrase pairs) as supplementary signal. The model trained on English data is evaluated zero-shot on the Hinglish test set without exposure to code-switched text during training.

3.3 Preprocessing Strategy

A key design decision is to apply approach-specific preprocessing, motivated by the different assumptions of each model. Table 2 summarizes the strategy.

4 Methods

4.1 Approach 1: Fine-Tuned MuRIL

We fine-tune `google/muril-base-cased` [4] directly on the Hinglish sarcasm training split. MuRIL was pre-trained on 17 Indian languages plus transliterations, making its tokenizer natively aware of Romanized Hindi tokens such as *kya*, *nahi*, and *waah*. A binary classification head is added on top of the `[CLS]` representation and trained with class-weighted cross-entropy loss (class weights: $w_{\text{sarcastic}} = 5.21$, $w_{\text{non-sarcastic}} = 0.55$) to address the significant label imbalance. Key configuration:

- **Base model:** `google/muril-base-cased`
- **Head:** Standard sequence classification head with dropout 0.1
- **Optimizer:** AdamW (standard Transformer defaults)
- **Training:** 3 epochs; final evaluation performed on the 10% test split.
- **Loss:** Class-weighted cross-entropy ($w_{\text{sarc.}} = 5.21$, $w_{\text{non-sarc.}} = 0.55$)
- **Performance:** The model achieved a Macro F1 of 0.8683 and an Accuracy of 94.67% on the test set.

MuRIL serves as the supervised upper bound for this study.

4.2 Approach 2: Zero-Shot Prompting with LLaMA-4

We evaluate Meta's LLaMA-4 Scout 17B [5] — a mixture-of-experts architecture with 17B active parameters out of 109B total — via the Groq API in a true zero-shot setting. No task-specific training or few-shot examples are used.

Prompt calibration. The initial prompt (v1) caused an **86% sarcasm prediction rate** (true rate: 9.6%), collapsing Macro F1 to ≈0.27. Root cause: the prompt provided only sarcastic cues with no base-rate signal, anchoring the model on the sarcastic class. The revised prompt (v2) explicitly (1) states that the vast majority of tweets are *NOT* sarcastic; (2) distinguishes sarcasm from anger and strong opinions; (3) instructs: *"When in doubt, output 0 (Non-Sarcastic)."* This corrected the prediction rate to ≈10%, matching the true base rate, and achieved Macro F1 = 0.7660 — a **40-point improvement from a single prompt revision**.

Implementation. `temperature=0.0` for reproducibility; 4 s inter-request delay (≈15 req/min); checkpoint every 25 predictions; exponential backoff on rate limits; failed predictions flagged as `-1` and never silently defaulted.

4.3 Approach 3: Cross-Lingual Transfer via XLM-RoBERTa

We train `xlm-roberta-large` [2] (560M parameters, 100-language pre-training) on English iSarcasmEval data and evaluate zero-shot on Hinglish. Training uses two phases:

Phase 1 — Classifier warmup. Backbone frozen; classification head trained for 1 epoch at lr = 5e-4.

Phase 2 — Full fine-tuning. All layers unfrozen; lr = 1e-5; 4 epochs; AdamW; gradient clipping max_norm = 1.0; class-weighted loss (mirroring iSarcasmEval, ≈15% sarcasm rate). Best checkpoint selected on English validation Macro F1.

5 Results

5.1 Main Results

Table 3 presents all results on the held-out test set. XLM-RoBERTa's pattern — 0.9048 accuracy but 0.4750 Macro F1 — is the signature

of majority-class collapse: a classifier that always predicts Non-Sarcastic achieves 0.9010 accuracy and 0.4739 Macro F1 on this split, confirming that Approach 3 detected zero sarcastic tweets.

5.2 Confusion Matrix Breakdown

Table 4 shows per-model confusion matrices. MuRIL achieves the most balanced FP/FN distribution (15 FP, 13 FN). LLaMA-4 exhibits a recall bias — catching most sarcasm (39 TP) at the cost of elevated false positives (43 FP). XLM-RoBERTa detects zero sarcastic instances.

5.3 LLaMA-4 Prompt Sensitivity

Table 5 quantifies the impact of prompt design. The shift from v1 to v2 is the single largest performance improvement in any component of this study — larger than the entire gap between MuRIL and LLaMA-4.

6 Analysis

6.1 Why MuRIL Outperforms All Others

MuRIL's dominance (Macro F1 = 0.8683) stems from three compounding advantages. **(1) Tokenization:** MuRIL's SentencePiece vocabulary was trained on 17 Indian languages including transliterated Hinglish. Tokens like *kya*, *nahi*, and *waah* map to meaningful subword representations rather than generic fragments. **(2) Pretraining distribution:** MuRIL's corpus includes Romanized Hindi, so its contextual representations natively encode Hinglish pragmatics. **(3) Supervised signal:** MuRIL is the only model that directly observes Hinglish sarcasm labels during training, learning the specific discourse markers — mock praise, ironic hashtags, exaggerated expressions — that distinguish sarcasm from genuine sentiment.

6.2 Why LLaMA-4 Is Competitive Despite Zero Training Data

LLaMA-4 (Macro F1 = 0.7660) sits only 10.2 F1 points below the fully supervised model. Two factors explain this. First, LLaMA-4's 17B active-parameter (109B total) instruction-tuned MoE architecture encodes broad multilingual pragmatic knowledge from web-scale pre-training, which includes substantial Hinglish content from Indian social media. Second, the calibrated prompt (v2) is critical: it anchors the model to the true base rate and explicitly separates sarcasm from anger, enabling the model's implicit pragmatic knowledge to activate correctly.

6.3 Why Cross-Lingual Transfer Failed

XLM-RoBERTa's zero-shot failure (Macro F1 = 0.4750, Sarcastic F1 = 0.000) is a complete majority-class collapse driven by four compounding factors:

- **Domain mismatch:** iSarcasmEval is English Twitter/Reddit. Sarcasm markers informative in English (formal hyperbole, ironic understatement) do not map to Hinglish markers (*Waah*, *kya baat hai*, mock praise in Roman script).
- **Vocabulary mismatch:** Code-switched token sequences fragment into generic subword tokens, losing semantic content.
- **Decision boundary miscalibration:** The boundary learned for English sarcasm (≈15% prior in iSarcasmEval) collapses on Hinglish with a 9.9% sarcasm rate under a completely different feature distribution.
- **Cultural pragmatic gap:** Hinglish sarcasm relies on Hindi-specific discourse markers and contextually grounded irony with no English structural equivalent.

This negative result is itself a contribution: it demonstrates that cross-lingual transfer of figurative language, unlike syntactic tasks, does not generalize to code-switched text without explicit domain adaptation.

6.4 Qualitative Error Analysis

Table 6 presents representative error cases for LLaMA-4. **False positives** arise primarily from *culturally intense non-sarcastic language* — patriotic expressions and emphatic affirmations that superficially resemble ironic overstatement. **False negatives** occur on *understated irony* conveyed through implicit contrast without any surface-level mock-praise marker. MuRIL, having learned the precise distributional signature of sarcasm in this dataset during training, handles both cases more robustly.

6.5 Statistical Reliability

With a test set of 525 examples (52 sarcastic), differences in Macro F1 may partially reflect sampling variance. We apply McNemar's test on the binary sarcasm predictions of MuRIL vs. LLaMA-4. The discordant pair counts (MuRIL correct, LLaMA-4 wrong: $n_{10} = 29$; LLaMA-4 correct, MuRIL wrong: $n_{01} = 9$) yield $\chi^2 = \frac{(|29-9|-1)^2}{29+9} = 9.74$, $p < 0.01$ (df = 1), confirming that MuRIL's advantage is statistically significant beyond sampling variation.

6.6 Comparative Paradigm Summary

Table 7 summarizes core trade-offs across all three paradigms.

7 Discussion

Three principal findings emerge from this study, alongside one notable data constraint. **Supervised fine-tuning on in-domain data is irreplaceable when labeled data is available.** MuRIL's 8.3-point advantage over LLaMA-4 and 39-point advantage over XLM-RoBERTa reflects both language-native tokenization and direct task supervision. The McNemar test ($p < 0.01$) confirms this gap is not attributable to sampling noise. **Well-calibrated zero-shot LLMs are competitive but fragile in low-resource settings.** LLaMA-4's Macro F1 = 0.7660 without any task-specific training is a strong result, yet the 40-point sensitivity to prompt calibration introduces practical fragility: uncalibrated deployment produces worse-than-baseline results. This sensitivity must be documented as a reproducibility concern for any zero-shot LLM evaluation. Future deployments should treat prompt base-rate calibration as a required engineering step, not an optional refinement. **Cross-lingual transfer of figurative language requires domain adaptation.** Unlike syntactic tasks where XLM-RoBERTa transfers reliably, sarcasm requires cultural and pragmatic knowledge not captured by surface cross-lingual representations. Even XLM-RoBERTa's full 100-language vocabulary, which includes Romanized Hindi tokens

Table 3: Main results on the Hinglish sarcasm test set (525 tweets; 52 sarcastic). All models evaluated on the identical stratified hold-out (`random_state=42`). Bold: best per column. †: majority-class baseline (always predict Non-Sarcastic).

Approach	Paradigm	Accuracy	Macro F1	Non-Sarc. F1	Sarcastic F1
Approach 1 — MuRIL	Supervised fine-tuning	**0.9467**	**0.8683**	**0.97**	**0.77**
Approach 2 — LLaMA-4 Scout	Zero-shot prompting	0.8971	0.7660	0.94	0.59
Approach 3 — XLM-RoBERTa	Cross-lingual transfer	0.9048	0.4750	0.9496	0.0000
Majority baseline†	Always predict Non-Sarc.	0.9010	0.4739	0.9481	0.0000

Table 4: Confusion matrix breakdown. TN = True Negative, FP = False Positive, FN = False Negative, TP = True Positive.

Model	TN	FP	FN	TP
MuRIL	460	15	13	37
LLaMA-4	432	43	11	39
XLM-RoBERTa	473	0	52	0

Table 5: LLaMA-4 prompt version comparison on the Hinglish test set.

Prompt	Pred. Sarc.	Macro F1	Sarc. F1
v1 (no base-rate)	≈86%	≈0.27	≈0.22
v2 (calibrated)	≈10%	0.7660	0.59
True base rate	9.9%	—	—

are insufficient; the learned decision boundary from English data is miscalibrated for Hinglish figurative language. Explicit domain adaptation or few-shot support is necessary before cross-lingual transfer can be practically deployed.

Finally, dataset availability constrained our planned script-level evaluation. While our initial proposal anticipated evaluating performance across Devanagari, Roman script, and transliterated Hindi, the Devanagari subset of the dataset was ultimately unavailable for use. Consequently, our analysis is strictly limited to Romanized and transliterated Hinglish. This constraint prevented us from testing whether the models struggle disproportionately with Romanized Hindi versus Devanagari script as originally intended.

8 Conclusion

We compared three paradigms for sarcasm detection on Hinglish code-switched social media text under identical experimental conditions. MuRIL (supervised fine-tuning) achieved the best Macro F1 of 0.8683, followed by LLaMA-4 zero-shot at 0.7660, and XLM-RoBERTa cross-lingual transfer at 0.4750, which collapsed to majority-class prediction. Prompt calibration is critical for zero-shot LLMs: a single poorly designed prompt reduced F1 by 40 points, and this fragility is statistically distinguishable from the supervised approach ($p < 0.01$, McNemar's test).A key limitation of this study was the unforeseen unavailability of the Devanagari dataset, which restricted our scope to Romanized text and precluded our proposed comparative analysis of script variations. Future work should explore: (1) few-shot prompting with Hinglish-specific sarcasm examples, to evaluate if few-shot examples improve detection of subtle Romanized irony; (2) domain-adaptive pre-training using unlabeled Hinglish data to bridge the XLM-RoBERTa gap; (3) ensemble approaches combining MuRIL's script-aware supervised precision with LLaMA-4's cultural pragmatic reasoning; (4) cultural context retrieval (RAG over Indian social media) to ground LLM inference on culturally intense non-sarcastic language; (5) extension to multi-class sentiment analysis beyond binary sarcasm detection; and (6) the procurement of comprehensive mixed-script datasets to effectively evaluate model performance across both Romanized and native Devanagari text.

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Table 6: Representative LLaMA-4 error cases. Tweets are transliterated/anonymized. Failure mode and likely cause identified per case.

Tweet (representative)	True	MuRIL	LLaMA-4	Failure cause
"Desh pe garv hai, yeh log sach mein kaam kar rahe hain!" [Proud of the country, these people are really doing good work!]	Non-Sarc.	Non-Sarc.	Sarc. (FP)	Culturally intense patriotic register misread as irony
"Aaj ka din bahut productive tha, office mein sab kuch normal tha." [Today was very productive, everything at the office was normal.]	Non-Sarc.	Non-Sarc.	Sarc. (FP)	Bland affirmation read as deadpan sarcasm
"Wah bhai, kya system hai. Form bhar diya, server down." [Wow bro, what a system. Filled the form, server down.]	Sarcastic	Sarc.	Sarc.	Both correct — explicit mock-praise marker *Wah*
"Interesting decision. Dekh lenge kya hota hai." [Interesting decision. We'll see what happens.]	Sarcastic	Sarc.	Non-Sarc. (FN)	Understated irony; no explicit mock-praise marker

Table 7: Comparative paradigm analysis: advantages, limitations, and root cause of performance differences.

Approach	Macro F1	Core Advantage	Core Limitation	Main Failure
MuRIL (supervised)	0.8683	Native Hinglish tokenization; direct task supervision; class-weighted loss	Requires labeled Hinglish data	Subtle irony without lexical cues
LLaMA-4 (zero-shot)	0.7660	Implicit multilingual pragmatics; no training; prompt calibration recovers base rate	Highly prompt-sensitive; rate-limited API	Culturally intense non-sarcasm (FP)
XLM-R (cross-lingual)	0.4750	Strong multilingual representations (100 languages, all scripts)	Majority-class collapse; English decision boundary fails on any Hinglish variant	All sarcasm missed (TP = 0)